

Addis Ababa University
Department of Computer Science
Complexity Theory (CoSc4112)
Course Outline

Course Title	Complexity Theory
Course Code	CoSc4112
Credit Hour:	3
Lecture:	3
Lab:	0
Prerequisite:	Formal Language Theory (Comp 571)
Course Objectives & Competences	By the end of this course, students will be able to: - Explain Models of Computation, resources (time and space), algorithms, computability, and complexity. - Understand Complexity classes, P/NP/PSPACE, reductions, hardness, completeness, hierarchy, relationships between complexity classes. - Learn Randomized computation and complexity; Logical characterizations, incompleteness and Approximability. - Understand Circuit complexity, lower bounds; Parallel computation and complexity; Counting problems and Interactive proofs. Learn Probabilistically
Course Description	The course focuses on the theory of computation. Topics include: Turing Machine(TM): StandardTM, Computability of TM, Construction of TMs; Computability: Overview of recursive function, primitive and partial recursive functions, recursive functions and enumerable languages, Turingcomputable; Undecidability: Turing decidable and Turing acceptable, undecidability problems: halting problem, some unsolvable problems about TMs, some unsolvable problems about CFLs; Computational complexity: Big-O notation, non-deterministic polynomial time, class P vs class NP, polynomial time reduction and NP-complete problems, Cook's theorem

Chapter 1: Turing Machines (TMs)

- Standard TM
- Computability of TM
- Techniques of TMs construction
- Church's hypothesis;

Chapter 2: Computability

- Introduction to recursive function theory; primitive and partial recursive functions
- Recursive and recursively enumerable languages
- Turing computable

Chapter 3: Undecidability

- Turing decidable and Turing acceptable
- Undecidable problems: TM halting problem; some unsolvable problem about TMs; some unsolvable problems about CFLs and more
- Undecidable problems.

Chapter 4: Computational Complexity

- Basics of algorithm analysis: Big O-notation
- Polynomial time and space; nondeterministic polynomial time; P vs NP
- Polynomial time reductions and NP-complete problems: Cook's theorem

Textbooks &References:

1. Peter Linz, An Introduction to Formal language Automata, *D.C. Heath*, 1990.
2. K.L.P Mishra, Theory of Computer Science, PHI, 2003.
3. John C.Martin, Introduction to Languages and the Theory of Computation, Tata McGraw-Hill, 2004